
TeamFour Track B

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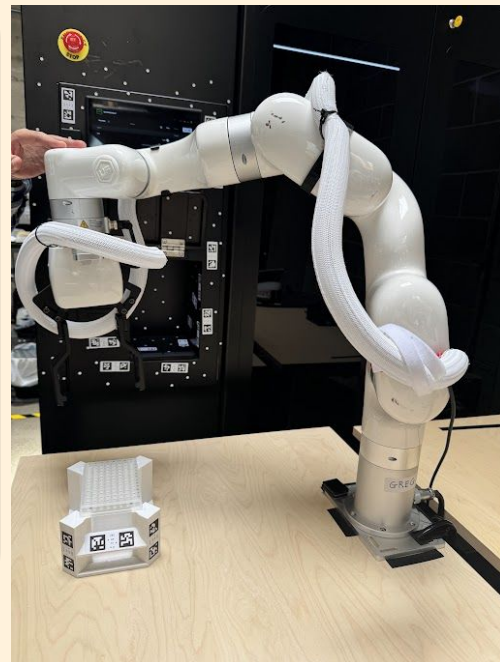
The Idea & Hypothesis

Idea

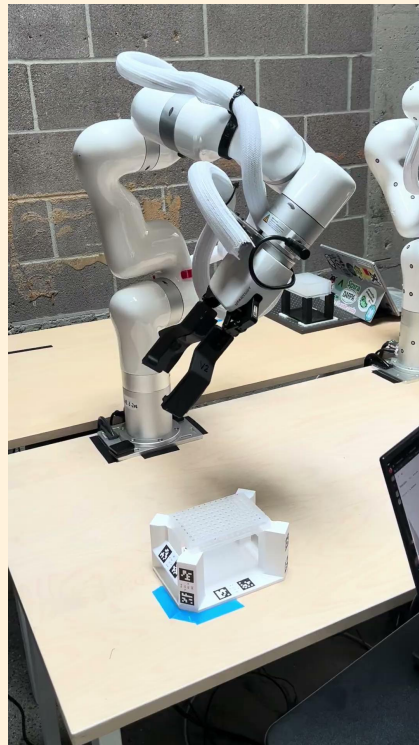
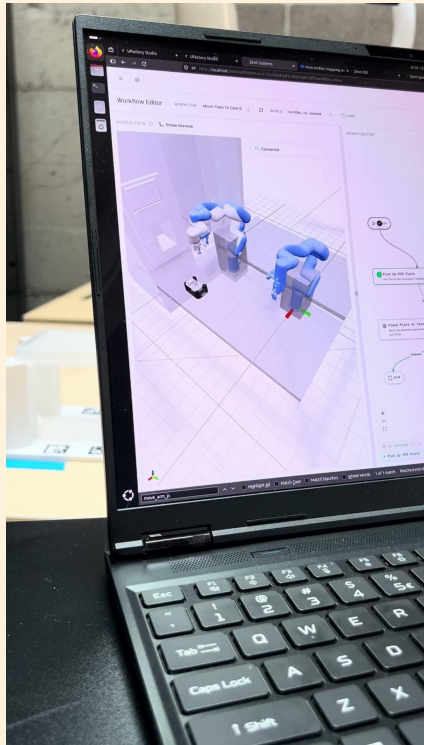
- Have the Zeon System recognize objects via an ArUco marker and dynamic gripping
- Place and store items in OpenShelf

Practical Utility

- Once an item is tagged, you can automate its storage with OpenShelf
- An agent can query and bring objects into the workcell as needed

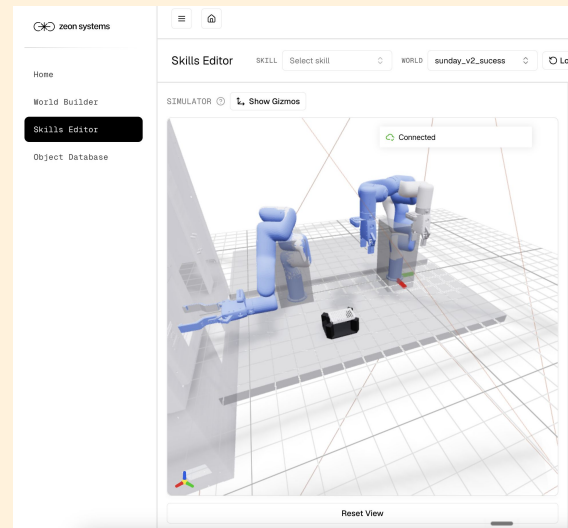


Demo Videos



Results

- Vibe coding is much harder than expected, even in simulation
 - Poor understanding of spatial images in CUA mode when running the actual simulation
 - Spent significant time getting used to the nuances of the Zeon System (e.g., using intermediate points to solve inverse kinematics)
- Sim-to-real gap
- We have a script that queries OpenShelf for the first open container of size $(155 \times 155 \times 275 \text{ mm})$ and then stores the item in that shelf



Next Steps

- Recalibrate the robotic arms and redefine anchor points as orientation hubs
- Better prompting and MCP servers in Zeon Cloud for coding agents
- Custom anchors on objects depending on Aruco marker
- OpenShelf handoff
- Real→Sim live

Thank you to

Zeon Systems
Bay Area Lab Automators
OpenShelf
SepiaBio
Google
Ufactory
SF Hardware meetup
Worldwide Studios

